In a Scheduled Jobs PR,[16] a new *Scheduled Jobs* page was added that adds visibility to pg_cron jobs. Scheduled Jobs that are scheduled and jobs that have run are displayed.

One final topic for maintenance is to learn about some helpful maintenance tooling.

Conducting Maintenance Tune-Ups

Performing database maintenance to keep operations optimal is part of your job as a database administrator.

The most important operations you've conducted are VACUUM, ANALYZE, and REINDEX (See: *Maintaining a PostgreSQL database health with ANALYZE, REINDEX, and VACUUM commands*[17]).

You also learned why and how to remove unused Indexes and duplicate or redundant indexes.

You saw the active_record_doctor[18] gem and database_consistency[19] gem; these can also be used to help with database maintenance. Database Consistency has various Checkers including a RedundantIndexChecker that helps you find redundant indexes to remove.

You also saw the rails-pg-extras[20] gem; it can be used for maintenance as well. This gem checks for Duplicate Indexes, Unused Indexes, and Bloat. The gem also looks for what it calls Null Indexes, which are indexes with a high proportion of NULL values. These may be good opportunities to replace a regular B-Tree index with a smaller Partial Index.

Performing maintenance regularly and monitoring the automatic maintenance will help you keep your database running smoothly.

Now you're ready to look at handling higher levels of concurrent activity.

16. https://github.com/andyatkinson/pghero/pull/3
17. https://andreigridnev.com/blog/2016-04-01-analyze-reindex-vacuum-in-postgresql/
18. https://github.com/gregnavis/active_record_doctor
19. https://github.com/djezzzl/database_consistency
20. https://github.com/pawurb/rails-pg-extras